

PROTEOMICS

Sensitive protein analysis with plexDIA

plexDIA increases throughput of proteomics analysis with low sample amounts.

High-throughput proteomic analysis of low-volume or single-cell samples continues to be limited in depth and throughput. Until now, throughput of mass spectrometry (MS)-based sensitive protein analysis has been increased either by chemical labeling or by simultaneous analysis of multiple peptides. The research group of Nikolai Slavov from Northeastern University and their collaborators have combined the two strategies to develop plexDIA.

Chemical labeling is typically used with data-dependent acquisition (DDA) strategies, but depth is limited by the number of precursors that can be analyzed at the MS2 level per run. Non-isobaric labels such as mTRAQ and dimethyl labeling increase the number of precursors, increasing analysis time for multiplexed DDA analysis. Isobaric labels such as tandem mass tags (TMTs), on the other hand, suffer from co-isolation

interference problems. A solution to this is data-independent acquisition (DIA), where multiple precursor ions are simultaneously analyzed in a label-free manner, and increasing the number of precursor ions does not increase analysis time.

To develop plexDIA, the researchers multiplexed samples labeled with non-isobaric isotopologous mass tags. “What’s very important to emphasize here is that these are not isobaric mass tags but non-isobaric, and that solves another problem, which is the co-isolation interference and the dynamic range compression that we suffered at least to some extent with isobaric mass tags. Now, with non-isobaric, we can do the quantification at MS1 and MS2 levels free of co-isolation interference,” explains Slavov.

plexDIA obtained a multiplicative increase in the rate of protein quantification. Using three amine-reactive non-isobaric

isotopologous mass tags, the researchers obtained quantification of threefold more protein samples while preserving proteome coverage and quantitative accuracy. “We have demonstrated very clearly the concept for n equals three with three non-isobaric mass tags, but we feel very confident that we can increase this to much higher plex,” says Slavov: “by optimizing the chemistry of the mass tags going forward, we can further increase the depth of proteome coverage and the ability to propagate sequences and most of all the throughput of analysis.”

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Research papers

Derks, J. et al. Increasing the throughput of sensitive proteomics by plexDIA. *Nat. Biotechnol.* <https://doi.org/10.1038/s41587-022-01389-w> (2022).